SKILL - 9

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SEC-8A

Problem Statement :

1. To Change Directories, Validate Paths, Update Working Directory, Handle Errors, Maintain Previous Directory, Test Navigation.

2. To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

Aim :

To implement directory navigation and a built-in command dispatcher in C, including path validation, working-directory updates, error handling, state maintenance, and in-process execution of built-in commands.

Procedure :

1. Define built-in functions for **pwd**, **cd**, and **exit**.

2. Create a dispatch table that maps each built-in command name to its corresponding function.

3. Read commands continuously using **fgets()** and remove the trailing newline.

4. Tokenize the command and optional path argument using **strtok()**.

5. Use **getcwd()** to identify the current working directory and **chdir()** to change directories.

6. Validate the requested path and report errors using **errno** and **strerror()** when navigation fails.

7. Record the previous directory and display the updated working directory after a successful directory change.

8. Search the dispatch table for the entered built-in command and execute the matching function directly within the process.

9. Report invalid commands without terminating the shell and maintain the current state until the **exit** command is issued.

Output :

```
dmvs@DMVS:/mnt/c/Testing/OSSP_Testing/skill$ ./Skill_9
=== Skill-9: Directory Navigation and Built-in Dispatch ===
Built-in commands: pwd, cd <path>, exit

shell> pwd
Current Directory: /mnt/c/Testing/OSSP_Testing/skill
shell> exit
Exiting shell...
dmvs@DMVS:/mnt/c/Testing/OSSP_Testing/skill$
```

The program starts with the Skill-9 banner and identifies **pwd**, **cd <path>**, and **exit** as built-in commands. The test shown executes **pwd** and reports the working directory as **/mnt/c/Testing/OSSP_Testing/skill**. The **exit** command then terminates the shell and prints the exit message.

Source Code - Skill_9.c (1/4):

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>
#include <errno.h>

typedef void (*builtin_func)(char *);

void builtin_pwd(char *arg);
void builtin_cd(char *arg);
void builtin_exit(char *arg);

struct builtin
{
    char *name;
    builtin_func function;
};

struct builtin dispatch_table[] =
{
    {"pwd", builtin_pwd},
    {"cd", builtin_cd},
    {"exit", builtin_exit}
};

int builtin_count = sizeof(dispatch_table) / sizeof(dispatch_table[0]);

void builtin_pwd(char *arg)
{
    char cwd[PATH_MAX];

    if (getcwd(cwd, sizeof(cwd)) != NULL)
        printf("Current Directory: %s\n", cwd);
    else
        perror("getcwd");
}
```

Source Code - Skill_9.c (2/4):

```c
void builtin_cd(char *arg)
{
    char previous[PATH_MAX];
    char current[PATH_MAX];

    if (getcwd(previous, sizeof(previous)) == NULL)
    {
        perror("getcwd");
        return;
    }

    if (arg == NULL || strlen(arg) == 0)
    {
        printf("cd: missing path\n");
        return;
    }

    if (chdir(arg) == -1)
    {
        printf("cd: %s: %s\n", arg, strerror(errno));
        return;
    }

    if (getcwd(current, sizeof(current)) != NULL)
    {
        printf("Previous Directory: %s\n", previous);
        printf("Working Directory Updated: %s\n", current);
    }
}

void builtin_exit(char *arg)
{
    printf("Exiting shell...\n");
    exit(0);
}

void execute_builtin(char *input)
```

Source Code - Skill_9.c (3/4):

```c
{
    char *command;
    char *argument;

    command = strtok(input, " \t");
    argument = strtok(NULL, "");

    if (command == NULL)
    {
        printf("Empty command.\n");
        return;
    }

    for (int i = 0; i < builtin_count; i++)
    {
        if (strcmp(command, dispatch_table[i].name) == 0)
        {
            dispatch_table[i].function(argument);
            return;
        }
    }

    printf("Invalid built-in command: %s\n", command);
}

int main()
{
    char input[256];

    printf("=== Skill-9: Directory Navigation and Built-in Dispatch ===\n");
    printf("Built-in commands: pwd, cd <path>, exit\n\n");

    while (1)
    {
        printf("shell> ");

        if (fgets(input, sizeof(input), stdin) == NULL)
```

Source Code - Skill_9.c (4/4):

```
            break;

        input[strcspn(input, "\n")] = '\0';

        execute_builtin(input);
    }

    return 0;
}
```

Observation and Analysis

Directory Navigation

The **builtin_cd()** function obtains the current directory before attempting navigation. It validates the supplied argument, calls **chdir()**, and reports a readable error when the requested path cannot be accessed. After a successful change, both the previous directory and the updated working directory are displayed.

Built-in Dispatch

The program uses a dispatch table containing the command names **pwd**, **cd**, and **exit** together with their function pointers. The entered command is compared with each table entry, and the corresponding function is invoked directly when a match is found.

State and Error Handling

Because the built-in commands execute in the shell process itself, a successful **cd** changes the shell's working-directory state. Invalid commands are handled by printing an error message rather than terminating the program. Missing paths are also detected before directory navigation is attempted.

Observed Test

The supplied execution demonstrates the command-dispatch mechanism through **pwd** and **exit**. The current directory is correctly identified as **/mnt/c/Testing/OSSP_Testing/skill**, after which the shell exits normally.

Result :

Skill-9 was successfully implemented in C. The program provides in-process built-in command dispatch and directory-navigation functionality with path validation, error handling, working-directory state maintenance, and controlled termination.